NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

One Result, Two Claims

AP Statistics · Unit 3 · Topic 3.13 · B: 2026-12-17 · E: 2027-01-29

[STAR] TODAY'S PROBLEM

Suppose a utility has 500,000 online accounts. It randomly selects 10,000, assigns 5,000 to message A and 5,000 to B, and records on-time payment. The pre-data question asks whether the population proportions differ at $\alpha = 0.05$; deployment requires A to improve payment by at least 5 percentage points.

Rowan: "I rejected equality, so A has proved an important improvement of at least 5 points. Deploy A."

Salma: "Rejecting equality supports a nonzero difference, but does not automatically establish the separate 5-point requirement."

On-time payment

Message	n	Paid	Other
A	5,000	3,050	1,950
B	5,000	2,950	2,050

FIRST TAKE — before the video (5 min)

Whose report is more defensible, Rowan's or Salma's? Cite the observed gap or the claim tested.

[BOOK] CALCULATE, INTERPRET, THEN BOUND THE CLAIM (keep on the page)

For $H_0 : p_1 - p_2 = 0$, pool $\hat{p}_c = (x_1 + x_2)/(n_1 + n_2)$ and use $z = \dfrac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c(1 - \hat{p}_c)(1/n_1 + 1/n_2)}}$. A two-sided test uses $P(|Z| \geq |z|)$. The p-value assumes H_0 is true and measures how often random chance would produce a difference at least as extreme. If $p \leq \alpha$, reject H_0; otherwise fail to reject it. A test against zero addresses whether a difference exists; it does not by itself establish that the difference exceeds a separate practical threshold.

Finish after the video:

DOK 1 **(a)** Calculate $\hat{p}_A$, $\hat{p}_B$, and the observed difference $\hat{p}_A - \hat{p}_B$.

DOK 2 **(b)** Verify conditions and carry out the two-sided two-sample z-test of $H_0 : p_A - p_B = 0$. Compute $\hat{p}_c$, pooled SE, z, and the p-value; interpret the p-value and decide at $\alpha = 0.05$.

DOK 3 **(c)** Decide which report is warranted. Use the p-value, observed gap, 5-point standard, and null value. Explain what the rejected report would mislead a reader to believe and name the confidence level and endpoint criterion needed for deployment.

Frame: Because $p =$ __________ is _________ α, I would ______________ H_0 and report ______________________________.

Frame: That decision does not establish 5 points because ___________________________________; deployment would require an interval ______________________________.

EXIT — turn this in

One thing the video changed about my first take: ____